



De La Salle University



Building an Integrated Financial Database Using SQL in R

FINSTAD Case 1

Section C01, Group 3

in partial fulfillment of the requirements for the course

FINSTAD

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Executive Summary

[Summarize the objectives, datasets used, major SQL findings, key statistical results, and investment recommendation here (Maximum 1 Page).]

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1. Introduction

[Discuss the background of the study, objectives of the analysis, and the importance of integrating multiple financial datasets.]

2. Data Sources

[Describe the selected U.S. asset, Philippine asset, cryptocurrency, SPY benchmark, and the four required FRED macroeconomic variables.]

3. Data Acquisition

[Present a table showing the data source, sample period, original observations downloaded, missing values, and duplicate observations for each dataset.]

Table 1: Data Acquisition Summary

Dataset	Source	Original Observations	Missing Values	Duplicates
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4. Data Cleaning

[Describe the cleaning procedures performed, treatment of missing values, duplicate observations, and return calculations.]

5. Database Integration

[Explain the SQL join strategy used. Include a table showing observations before joins, after joins, and in the final dataset. Justify the strategy.]

Table 2: Database Integration Summary

Dataset	Original Observations	After Joins	Final Analytical Dataset
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6. SQL-Based Financial Investigation

[Present at least TEN (10) financial SQL investigations containing Objective, SQL/dplyr Code, Output, and Interpretation.]

6.1. SQL Investigation 1

Objective: [Objective]

Code: [Code]

Output: [Output]

Financial Interpretation: [Interpretation]

7. Descriptive Statistical Analysis

[Present mean, median, variance, standard deviation, minimum, maximum, range, skewness, kurtosis and discuss implications.]

8. Data Visualization

[Present at least EIGHT (8) professional-quality figures with titles, numbers, and interpretation.]

9. Exploratory Financial Analysis

[Answer financial questions using evidence from SQL investigations, stats, and visualizations (compare returns, volatility, inflation, high-VIX, interest-rates).]

10. Investment Recommendation

[Recommend allocation of USD 1,000,000 among U.S., PH, crypto, and cash. Justify using evidence.]

11. Executive Dashboard

[Include the 1-page executive dashboard with key stats, 3-5 charts, SQL findings, and recommendation.]

12. Conclusion

[Summarize findings, limitations, and recommendations for future work.]

A. Appendix A - Complete R Code

[Include the complete R script used in the analysis.]

B. Appendix B - Additional Tables and Figures

[Include supplementary outputs that are too large for the main report.]